PRACTICAL 2

IOC VALIDATION USING THREAT INTELLIGENCE

1. Objective
The objective of this step is to validate multiple Indicators of Compromise (IOCs) using a threat intelligence platform to determine their reputation status (clean, suspicious, or malicious) and understand their role in SOC analysis.

Tool Used
IOC validation was performed using:
- VirusTotal

VirusTotal aggregates results from multiple security vendors to provide detection, reputation, and classification of IP addresses.

3. Implementation
Three IP addresses were analyzed:
- 8.8.8.8 (Public DNS - Google)
- 45.33.32.156 (External hosted IP)
- 192.168.1.100 (Internal/private simulated IP)

Each IP was searched in VirusTotal, and detection results from multiple security vendors were reviewed.

4. Observations
8.8.8.8
- Detected as a **clean and trusted IP**
- Belongs to Google LLC
- No malicious activity detected across vendors
- Used as baseline trusted infrastructure

45.33.32.156
- **4/94 security vendors flagged this IP as malicious**
- Classified as suspicious by multiple threat intelligence sources
- Indicates possible malicious or abusive activity
- Requires monitoring in SOC environments

192.168.1.100
- Private IP address used in simulated environment
- Used for internal correlation with login activity and network behavior
- Not directly analyzed for public reputation but included for contextual SOC correlation

5. Result Summary

IP Address	Type	Detection Result	Status
8.8.8.8	Public DNS	0/94 detections	Clean
45.33.32.156	External IP	4/94 detections	Malicious/Suspicious
192.168.1.100	Internal IP	N/A	Simulated/Internal

6. Interpretation

The analysis demonstrates how SOC analysts classify IP addresses using threat intelligence platforms. Trusted infrastructure (8.8.8.8) is separated from potentially malicious external IPs (45.33.32.156), while internal IPs are used for correlation within organizational logs.
This step highlights the importance of IOC validation before using data for alert enrichment and threat hunting.

7. SOC Relevance

This step contributes to SOC workflow by enabling:

- IOC validation using threat intelligence
- Differentiation between clean and malicious IPs
- Baseline establishment for network behavior
- Input for alert enrichment and correlation

9. Conclusion

The IOC validation process successfully identified trusted and potentially malicious IP addresses using VirusTotal. This step forms the foundation of threat intelligence integration in SOC operations and supports further enrichment and threat hunting activities.